
# Mechatronic Sensors Trainer

## Fundamental Labs – Passive Infrared

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Caution

**This equipment is designed to be used for educational and research purposes and is not intended for use by the public.** The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

# Passive Infrared Lab

## Why Explore Passive Infrared Sensors?

When designing a mechatronics application, an important question to consider is whether a passive or active sensor is required. Passive sensors are desirable when there are mechanical and electrical constraints such as size, power draw, mechanical complexity, etc. Passive Infrared sensors offer a simple way of converting infrared emissions into a voltage measurement. Passive Infrared sensors are used in a variety of systems such as warehouse security systems, automatic lights, automatic doors, etc.

## Background

This lab is part of the Fundamentals labs for the Mechatronic Sensors Trainer. These labs will guide you through the sensors that are part of the board and the ones that are provided as external sensors so you can understand how to read from the sensors, what they read, when they are used, and you feel comfortable using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Mechatronics/**018_passive_infrared**.

## Getting Started

In this lab, you will use the **Sensors Trainer** to learn about the basics of **Passive Infrared** sensors. The lab will guide you through environmental considerations and measurement interpretation for Passive Infrared sensors.

Before you begin this lab, ensure that the following criteria are met:

- Make sure the Sensors Trainer has been setup and tested. See the Quick Start Guide (Quanser/2_quick_start_guides/sensors_trainer) for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
    - If using Python, we recommend using Visual Studio Code.